A new primal dual method for fully nonconvex composite optimization

School of Mathematical Sciences, Beihang University, Beijing 100191, China.

Luxuan Li

3 Dec, 2024

CONTENT

1 Introduction

2 Method

3 Main Results

4 Discussion

Introduction

- **Composite nonconvex nonsmooth finite-sum optimization problem**

$$\min_{x \in \mathcal{X}} f(x) + h(g(x))$$

  - $\mathbb{X}$, $\mathbb{B}$ and $\mathbb{Y}$ are finite-dimentional vector spaces.
  - $f: \mathbb{X} \to \mathbb{R}$ is a continuously differentiable possibly nonconvex, and may have a finite-sum structure

$$f = \frac{1}{n} \sum_{i=1}^{n} f_i(x)$$

  - $g = \big(g_1(x), g_2(x), \cdots, g_m(x)\big)$ is a nonlinear operator. Each $g_i(x): \mathbb{B} \to \mathbb{R}$ is continuously differentiable.
  - $h: \mathbb{Y} \to (-\infty, +\infty]$ is a simple, proper and nonsmooth, possibly nonconvex.
  - n can be large.

- Challenges

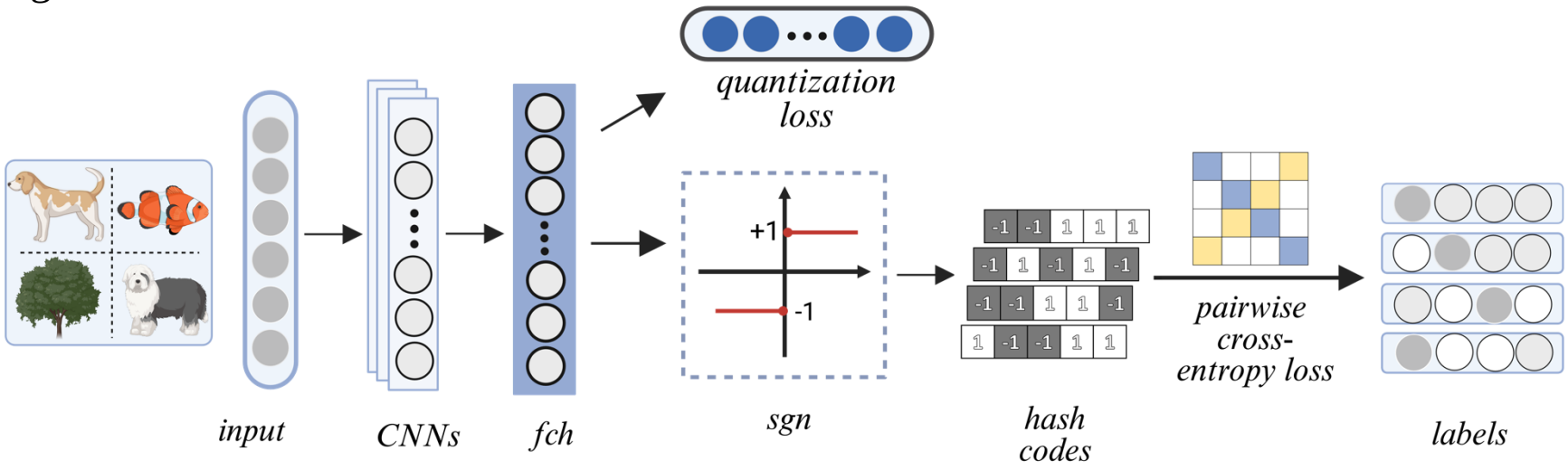
  - Nonconvexity of the problem
  - The coupling between a nonlinear operator and a nonsmooth function
  - The inefficiency of computing the proximal operator of $h \circ g$

Introduction

- Deep pairwise supervised hashing

$$\min_{\boldsymbol{\theta}\in\mathcal{X}} \frac{1}{n}\sum_{i=1}^{n} f_i(\boldsymbol{\theta}) + \sum_{i=1}^{n} h(tanh(D(\boldsymbol{\theta};\boldsymbol{a}_i)))$$

- f_i is the loss function, preserving the similarity between hamming space and input space.
- $h: \mathbb{Y} \rightarrow (-\infty, +\infty]$ is the binary hash regularization, which usually has the binarization approximation properties at -1 and 1.
- n can be large.



Method

- Primal problem:

$$\min_{x \in \boldsymbol{\mathcal{X}}} f(x) + h(g(x)) \qquad \text{(P)}$$

$$f^*(y) := \sup_{x \in \mathbb{X}} \{\langle y, x \rangle - f(x)\}, \;\; y \in \mathbb{X}^*$$

- Dual problem:

$$\max_{y \in \mathcal{Y}} \min_{x \in \mathcal{X}} L(x, y) = f(x) + \langle y, g(x) \rangle - h^*(y) \qquad \text{(D)}$$

Method

- For the update of x, the nonlinear opeator $g(x)$ is linearized at x^k, and J(x) is the Jacobian matrix;
- For the update of y, , the nonlinear opeator $g(x)$ is linearized at x^{k+1}.

$$
\begin{aligned}
x^{k+1} &= \arg\min_x \left\{ \langle \nabla f(x^k), x - x^k \rangle + \langle y^k, g(x^k) + J(x^k)(x - x^k) \rangle + \frac{1}{2\tau} \|x - x^k\|^2 \right\} \\
&= \arg\min_x \left\{ \langle \nabla f(x^k) + J(x^k)^T y^k, x - x^k \rangle + \frac{1}{2\tau} \|x - x^k\|^2 \right\} \\
\tilde{x}^{k+1} &= 2x^{k+1} - x^k \\
y^{k+1} &= \arg\min_y \left\{ h^*(y) - \langle g(x^{k+1}) + J(x^{k+1})(\tilde{x}^{k+1} - x^{k+1}), y \rangle + \frac{1}{2} \|y - y^k\|_{M_k}^2 \right\} \\
&= \arg\min_y \left\{ h^*(y) - \langle g^{k+1}(\tilde{x}^{k+1})), y \rangle + \frac{1}{2} \|y - y^k\|_{M_k}^2 \right\}
\end{aligned}
$$

where $M_k = \tau J(x^k) J(x^k)^T$

Main Results

- Main Assumptions

Assumption 1 *1. f is L_f-Lipschitz smooth;*

2. $\forall i = 1, 2, \cdots, m$, $g_i(x)$ is $L_{g_i}^0$-Lipschitz continuous: $\|\nabla g_i(x)\| \leq L_{g_i}^0$;
$\nabla g_i(x)$ L_{g_i}-Lipchitz continuous: $H_{g_i}(x) \preceq L_{g_i} I$, i.e., $\lambda_{max}(H_{g_i}(x)) \leq L_{g_i}, \forall x \in \mathcal{X}$

3. $J(x)J(x)^T$ is positive defined for $x \in \mathcal{X}$

Here, we obtain:

1. $g(x) : \mathbb{R}^n \to \mathbb{R}^m = (g_1(x), \cdots, g_m(x))^T$ L_g^0-Lipchitz continuous, $L_g^0 := \sum_{i=1}^m L_{g_i}$

$$\|g(x) - g(x')\| \leq L_g^0 \|x - x'\|$$

2. $J(x)$ has a upper bound: $\forall x \in \mathcal{X}$

$$\|J(x)\| \leq \sqrt{\sum_{i=1}^m (L_{g_i}^0)^2} = M_g^0$$

Main Results

Define the Lyapunov function

$$\mathcal{L}(x, y, u, v) = L(x^k, y^k) - a\|x - u\|^2 + b\|x - v\|^2,$$

where a , b , c satisfy

$$\begin{cases} a & = \frac{2\delta}{\tau^2} (> 0) \\ b & = \frac{1}{2\tau} - \frac{\delta}{\tau^2} - \frac{L_f}{2} - \frac{ML_g^0}{2} - c_2 (> 0) \\ c & = b - c_1 (> 0) \\ c_1 & = L_f^2 + \sum_{i=1}^m \left(\tau M L_{g_i}^0 L_{g_i}\right)^2 \\ c_2 & = \delta M^2 \left(\sum_{i=1}^m L_{g_i}^2\right) + 2\delta(\frac{1}{\tau} + L_f)^2 \end{cases}$$

for some $\delta > 0$.

Main Results

Lemma 3 (sufficient descent) *Let us define* $z^k = \{x^{k+1}, x^k, x^{k-1}, y^k\}$, $u^k = \{x^k, x^{k-1}\}$. *Under the above assumptions, with the stepsize* τ *set to satisfy*

$$\begin{cases} \frac{1}{\tau} & > \frac{L_f}{2} + \frac{ML_g^0}{2} + c_2 \\ \frac{1}{2\tau} - \frac{\delta}{\tau^2} & > \frac{L_f}{2} + \frac{ML_g^0}{2} + c_1 + c_2, \end{cases}$$

$\{\mathcal{L}(z^k)\}_{k \in N}$ *is non-increasing, and the following relation holds*

$$\mathcal{L}(z^{k+1}) - \mathcal{L}(z^k) \leq c\|u^{k+1} - u^k\|^2 \tag{1.2}$$

Main Results

Lemma 4 (subgradient boundedness) *Let us define* $\kappa_1 = \frac{24}{\tau^2} + 8(M_g^0)^2$,
$\kappa_2 = 3\left(L_f^2 + \frac{1}{\tau^2} + 6M^2\sum_{i=1}^m L_{g_i}^2 + 8\left(\frac{1}{\tau} + L_f\right)^2\right)$,
$\kappa_3 = 2(M_g^0)^2\left(1 + \tau^2\left(2M^2\sum_{i=1}^m L_{g_i}^2 + 4\left(\frac{1}{\tau} + L_f\right)^2\right)\right)$,
$\kappa = \max\{\kappa_1, \kappa_2 + \kappa_3\}$. *Then, the following holds:*

$$\begin{aligned}\operatorname{dist}(0, \partial L(x^k, y^k))^2 &\leq \kappa_1\|x^{k+1} - x^k\|^2 + (\kappa_2 + \kappa_3)\|x^k - x^{k-1}\|^2 \\ &\leq \kappa\|u^{k+1} - u^k\|^2\end{aligned} \tag{1.3}$$

Discussion

- 假设可否减弱?
- 是否可以做随机及加速推广?
- 数值效果暂时未知。